Assessment Menu: Teacher Guide

The Equation Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
SOLVE	Solve multi-step equations	A1, A2, E1, E6, S1
DISTRIB	Distribute and combine like terms	A3, A4, E2, S2
LITERAL	Solve literal equations (formulas)	A5, E3, E4, S3
MODEL	Write equations from words	A6, E5, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 SOLVE	Solve: $3x - 7 = 11$	**Answer:** $x = 6$ Add 7 to both sides: $3x = 18$ Divide both sides by 3: $x = 6$	2 pts: Adds 7 to both sides 1 pts: Divides by 3 2 pts: Correct answer $x = 6$
A2 SOLVE	Solve: $5x + 4 = 2x + 19$	**Answer:** $x = 5$ Subtract 2x from both sides: $3x + 4 = 19$ Subtract 4: $3x = 15$ Divide by 3: $x = 5$	2 pts: Moves the variable terms to one side 1 pts: Isolates $3x = 15$ 2 pts: Correct answer $x = 5$
A3 DISTRIB	Simplify: $4(2x - 3) + 5x$	**Answer:** $13x - 12$ Distribute the 4: $8x - 12 + 5x$ Combine like terms: $13x - 12$	2 pts: Distributes 4 to both terms ($8x$ and -12) 1 pts: Combines $8x + 5x$ 2 pts: Correct answer $13x - 12$
A4 DISTRIB	Solve: $3(x + 4) = 21$	**Answer:** $x = 3$ Distribute: $3x + 12 = 21$ Subtract 12: $3x = 9$ Divide by 3: $x = 3$	2 pts: Distributes 3 to both terms 1 pts: Subtracts 12 2 pts: Correct answer $x = 3$
A5 LITERAL	Solve for y: $2x + y = 9$	**Answer:** $y = 9 - 2x$ Subtract 2x from both sides: $y = 9 - 2x$	3 pts: Subtracts 2x from both sides 2 pts: Correct answer $y = 9 - 2x$ (or $-2x + 9$)
A6 MODEL	A plumber charges a $40 visit fee plus $25 per hour. Write an equation for a $140 bill and solve it for the hours h.	**Answer:** $40 + 25h = 140$, so $h = 4$ hours. Fee plus hourly charge equals the bill: $40 + 25h = 140$ Subtract 40: $25h = 100$ Divide by 25: $h = 4$	2 pts: Correct equation $40 + 25h = 140$ 2 pts: Solves correctly 1 pts: States $h = 4$ hours

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 SOLVE	Solve: $4(2x - 1) - 3 = 5(x + 2) + 1$	**Answer:** x = 6 Distribute: $8x - 4 - 3 = 5x + 10 + 1$ Combine: $8x - 7 = 5x + 11$ Subtract 5x and add 7: $3x = 18$ Divide by 3: x = 6	3 pts: Distributes both 4 and 5 correctly 2 pts: Combines like terms on each side 2 pts: Collects variable terms and constants correctly 3 pts: Correct answer x = 6
E2 DISTRIB	Simplify $3(2x - 5) - 2(x - 4) + 7$ completely, then evaluate your result when $x = -3$.	**Answer:** 4x; at x = −3 the value is −12. Distribute 3 and −2: $6x - 15 - 2x + 8 + 7$ Combine x-terms: $6x - 2x = 4x$ Combine constants: $-15 + 8 + 7 = 0$ Result: 4x. Substitute x = −3: $4(-3) = -12$	3 pts: Distributes 3 correctly 3 pts: Distributes −2 correctly (+8, not −8) 2 pts: Simplifies to 4x 2 pts: Evaluates to −12
E3 LITERAL	The area of a trapezoid is $A = \frac{h(b + c)}{2}$. **(a)** Solve the formula for *b*. **(b)** Find *b* when A = 42, h = 6, and c = 5.	**Answer:** (a) $b = \frac{2A}{h} - c$ (b) b = 9 (a) Multiply both sides by 2: $2A = h(b + c)$ Divide both sides by h: $\frac{2A}{h} = b + c$ Subtract c: $b = \frac{2A}{h} - c$ (b) $b = \frac{2(42)}{6} - 5 = 14 - 5 = 9$	3 pts: Multiplies both sides by 2 3 pts: Divides by h and subtracts c 2 pts: Substitutes correctly in part (b) 2 pts: Correct b = 9
E4 LITERAL	Solve for *x*: $3(x - a) = 2x + b$. Then find *x* when a = 4 and b = −5.	**Answer:** x = b + 3a; when a = 4 and b = −5, x = 7. Distribute: $3x - 3a = 2x + b$ Subtract 2x: $x - 3a = b$ Add 3a: $x = b + 3a$ Substitute: $x = -5 + 3(4) = 7$	3 pts: Distributes 3 to both terms 3 pts: Gets x alone: x = b + 3a 2 pts: Substitutes correctly 2 pts: Correct x = 7
E5 MODEL	Gym A charges a $20 joining fee plus $15 per month. Gym B charges $50 plus $10 per month. Write and solve an equation for the number of months *m* when the total costs are equal. What is that total cost?	**Answer:** 20 + 15m = 50 + 10m, so m = 6 months; each costs $110. Set the total costs equal: $20 + 15m = 50 + 10m$ Subtract 10m: $20 + 5m = 50$ Subtract 20: $5m = 30$, so m = 6 Cost: $20 + 15(6) = 110$ (check: $50 + 10(6) = 110$)	3 pts: Correct equation $20 + 15m = 50 + 10m$ 3 pts: Solves correctly for m 2 pts: m = 6 months 2 pts: Total cost $110
E6 SOLVE	Dev solved $2(3x - 4) = 4x + 6$ like this: $6x - 4 = 4x + 6$ $2x = 10$ $x = 5$ Find his mistake, explain it, and give the correct solution.	**Answer:** He multiplied only the 3x by 2 and left the −4. Correct solution: x = 7. Mistake: the 2 must multiply **both** terms inside the parentheses: $2(3x - 4) = 6x - 8$, not $6x - 4$. Correct: $6x - 8 = 4x + 6$ Subtract 4x and add 8: $2x = 14$ Divide by 2: x = 7. Check: $2(21 - 4) = 34$ and $4(7) + 6 = 34$ ✓	4 pts: Identifies the mistake: 2 was not distributed to −4 3 pts: Explains why in a complete sentence 3 pts: Correct solution x = 7

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 SOLVE	Mia thinks of a number n, multiplies it by 4, subtracts 6, divides the result by 2, and then adds 5. She gets 20. **(a)** Write an equation for her steps. **(b)** Solve it to find n. **(c)** Check by following her steps forward.	**Answer:** (a) $\frac{4n-6}{2}+5=20$ (b) n = 9 (c) 36 → 30 → 15 → 20 ✓ (a) $\frac{4n-6}{2}+5=20$ (b) Work backward. Subtract 5: $\frac{4n-6}{2}=15$ Multiply by 2: $4n-6=30$ Add 6: $4n=36$, so $n=9$ (c) $9\times4=36$; $36-6=30$; $30\div2=15$; $15+5=20$ ✓	6 pts: Correct equation (2 pts each: multiply/subtract, divide the whole numerator by 2, add 5) 6 pts: Solves with correct, shown steps 4 pts: Correct n = 9 4 pts: Checks by running the steps forward to 20
S2 DISTRIB	Lena solved $3(x-2)-2(x-5)=4x+1$: $3x-6-2x-10=4x+1$ $x-16=4x+1$ $-3x=17,\ x=-\frac{17}{3}$ **(a)** Find and explain her mistake. **(b)** Solve correctly. **(c)** Check in the original equation.	**Answer:** (a) She wrote $-2(-5)$ as -10; it is $+10$. (b) $x=1$ (c) both sides equal 5. (a) The -2 multiplies both terms: $-2(x-5)=-2x$ **+ 10**. A negative times a negative is positive. (b) $3x-6-2x+10=4x+1$ Combine: $x+4=4x+1$ Subtract x and 1: $3=3x$, so $x=1$ (c) Left: $3(1-2)-2(1-5)=-3+8=5$. Right: $4(1)+1=5$ ✓	5 pts: Identifies the sign error: $-2\cdot(-5)=+10$ 3 pts: Explains why in a complete sentence 6 pts: Solves correctly from the corrected line ($x+4=4x+1$) 3 pts: Correct $x=1$ 3 pts: Checks in the original equation (both sides 5)
S3 LITERAL	The balance of an account with simple interest is $A=P+Prt$. **(a)** Solve for t. **(b)** Solve for P. (Hint: factor first.) **(c)** Find P when A = 1,260, r = 0.05, and t = 4.	**Answer:** (a) $t=\frac{A-P}{Pr}$ (b) $P=\frac{A}{1+rt}$ (c) P = 1,050 (a) Subtract P: $A-P=Prt$. Divide by Pr: $t=\frac{A-P}{Pr}$ (b) Factor: $A=P(1+rt)$ Divide by $(1+rt)$: $P=\frac{A}{1+rt}$ (c) $1+rt=1+(0.05)(4)=1.2$, so $P=\frac{1{,}260}{1.2}=1{,}050$	4 pts: Part (a): subtracts P and divides by Pr 4 pts: Part (b): factors $A=P(1+rt)$ 4 pts: Part (b): divides by $(1+rt)$ 4 pts: Part (c): computes $1+rt=1.2$ 4 pts: Part (c): correct P = 1,050
S4 MODEL	A rectangle’s length is 4 cm more than 3 times its width. Its perimeter is 72 cm. **(a)** Define a variable and write an equation. **(b)** Find the length and width. **(c)** A square has the same perimeter. Which shape has the greater area, and by how much?	**Answer:** (a) $2(3w+4)+2w=72$ (b) w = 8 cm, length 28 cm (c) The square: $324-224=100$ cm². (a) Let w = width, so length = $3w+4$: $2(3w+4)+2w=72$ (b) Distribute: $6w+8+2w=72$, so $8w=64$ and $w=8$. Length $=3(8)+4=28$. (c) Rectangle area: $8\times28=224$ cm². Square side: $72\div4=18$, area $18\times18=324$ cm². The square is larger by $324-224=$ **100 cm²**.	5 pts: Defines the variable and writes a correct equation 5 pts: Solves: w = 8 and length 28 4 pts: Rectangle area 224 and square side 18 3 pts: Square area 324 3 pts: Compares: the square is larger by 100 cm²